Masaki Ogura

Curriculum Vitae

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School of Engineering Building, Office C1-233,
1-4-1 Kagamiyama, Higashi-Hiroshima,
Hiroshima 739-8525, Japan
oguram@hiroshima-u.ac.jp
https://csslab.jp/en/

AREAS OF EXPERTISE

Control theory, network science, optimization, stochastic processes, biological physics

EDUCATION

- Ph.D. (Mathematics), *Texas Tech University*. 2014
- M.Sc. (Informatics), *Kyoto University*. 2009
- B.Eng., *Kyoto University*. 2007

PROFESSIONAL APPOINTMENTS

- Professor, Graduate School of Advanced Science and Engineering, Hiroshima University, Japan. Apr 2024–present
- Visiting Professor, Department of Bioinformatic Engineering, Graduate School of Information Science and Technology, Osaka University, Japan. Apr 2024–present
- Associate Professor, Department of Bioinformatic Engineering, Graduate School of Information Science and Technology, Osaka University, Japan. Nov 2019–Mar 2024
- Visiting Associate Professor, Division of Information Science, Graduate School of Science and Technology, Nara Institute of Science and Technology, Japan. Apr 2020–Mar 2021
- Assistant Professor, Graduate School of Information Science, Nara Institute of Science and Technology, Japan. Mar 2017–Oct 2019
- Postdoctoral Researcher, Department of Electrical and Systems Engineering, University of Pennsylvania, Oct 2014–Feb 2017

SHORT TERM VISITS

- Department of Mechanical Engineering, University of Hong Kong. 2018, 2019
- ICTEAM Institute, Université catholique de Louvain, Belgium. 2013

SELECTED AWARDS AND HONORS

- Runner-up of the 2019 Best Paper Award, *IEEE Transactions on Network Science and Engineering*. 2019
- Summer Dissertation/Thesis Research Award, Texas Tech University. 2014
- Cash Family Endowed Fellowship, Texas Tech University. 2013

- Best Paper Award, The Society of Instrument and Control Engineers. 2012

PUBLICATIONS

Book Chapters

[4] K. Sakurama, K. Kashima, T. Ikeda, N. Hayashi, K. Hoshino, M. Ogura, and C. Zhao, "System-Control-Based Approach to Car-Sharing Systems," in *Advanced Mathematical Science for Mobility Society*. Springer Singapore, 2024, pp. 127–171.

[3] V. M. Preciado, M. Zargham, C. Nowzari, S. Han, M. Ogura, A. Jadbabaie, and G. J. Pappas, "Bio-inspired Framework for Allocation of Protection Resources in Cyber-Physical Networks," in *Principles of Cyber-Physical Systems*. Cambridge University Press, 2020, pp. 293–322.

[2] M. Ogura and V. M. Preciado, "Optimal Containment of Epidemics in Temporal and Adaptive Networks," in *Temporal Networks Epidemiology*. Springer, 2017, pp. 241–266.

[1] M. Ogura and C. F. Martin, "Linear Switching Systems and Random Products of Matrices," in *Mathematical System Theory – Festschrift in Honor of Uwe Helmke on the Occasion of his Sixtieth Birthday*, K. Hüper and J. Trumpf, Eds. CreateSpace, 2013, pp. 291–300.

Refereed Journal Articles

[74] Y. Chen, H. Zhao, M. Ogura, Y. Gao, and L. Peng, "Data-driven dual-channel dynamic event-triggered load frequency control for multiarea power systems under DoS attacks," *Energy Science & Engineering*, vol. 14, no. 2, pp. 780–792, 2026.

[73] Q. Shi, M. Ogura, Q. Wang, F. Cheng, and L. Peng, "Leading cruise control under human reaction delays: stabilizability, string stability, and mixed H_2/H_∞ synthesis," *IEEE Transactions on Automation Science and Engineering* (accepted for publication), 2026.

[72] Y. Higuchi, R. Nagai, Y. Bai, A. Okazaki, N. Wakamiya, and M. Ogura, "Development and evaluation of a black-box optimization framework for weather-intervention design," *Journal of Computational Science*, vol. 97, p. 102850, 2026.

[71] H. Maeda, Y. Bai, N. Wakamiya, and M. Ogura, "Modeling and analysis of locust hoppers dynamics under olfactory receptor 35 deficiency," *Ecological Modelling*, vol. 517, p. 111608, 2026.

[70] S. Yang, L. Peng, M. Ogura, and Y. Chen, "Dual-channel dynamic event-triggered data-driven control for energy storage system using quantised information," *Journal of Control and Decision* (accepted for publication), 2026.

[69] Q. Wang, Y. Bai, M. Ogura, and L. Peng, "Periodic event-triggered generalized dissipative state estimation for hidden semi-Markov jump systems," *ISA Transactions*, vol. 173, pp. 122–130, 2026.

[68] S. Yang, L. Peng, M. Ogura, Y. Gao, and Y. Chen, "Dual-channel event-triggered data-driven control for energy storage system with input saturation and packet loss," *Journal of Automation and Intelligence* (accepted for publication), 2026.

[67] Y. Chen, H. Zhao, M. Ogura, H. Yu, and L. Peng, "Data-driven event-triggered fixed-time load frequency control for multi-area power systems with input delays," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 72, no. 12, pp. 8492–8504, 2025.

[66] Y. Chen, H. Zhao, M. Ogura, Y. Gao, and L. Peng, "Data-driven dual-channel dynamic event-triggered load frequency control for multi-area power systems with uniform quantizer," *Science Progress*, vol. 108, no. 4, p. 00368504251391464, 2025.

[65] Y. Deng, M. Ogura, Y. Bai, Y. Wang, and M. Feroskhan, "Safety-critical control for nonlinear systems with complex input constraints," *IEEE Transactions on Automatic Control*, vol. 70, no. 10, pp. 7016–7023, 2025.

[64] S. Yang, H. Zhao, M. Ogura, and L. Peng, "Data-driven adaptive optimal output tracking with assured convergence rate," *Journal of the Franklin Institute*, vol. 362, no. 16, p. 108086, 2025.

[63] F. Sato, M. Ogura, A. Sashie, Y. Bai, M. Shimono, and N. Wakamiya, "Neural SDE-based spike control of noisy neurons," *PLOS One*, vol. 20, no. 9, p. e0330607, 2025.

[62] R. Nagai, Y. Bai, M. Ogura, S. Kotsuki, and N. Wakamiya, "Evaluation of effectiveness of intervention strategy in control simulation experiment through comparison with model predictive control," *Nonlinear Processes in Geophysics*, vol. 32, no. 3, pp. 281–292, 2025.

[61] T. Ohtsuka, A. Okazaki, M. Ogura, and S. Kotsuki, "Convex optimization of initial perturbations toward quantitative weather control," *Scientific Online Letters on the Atmosphere*, vol. 21, pp. 2025–020, 2025.

[60] A. Fujioka, M. Ogura, and N. Wakamiya, "Cyclic pursuit formation control for arbitrary desired shapes," *Journal of the Franklin Institute*, vol. 362, no. 2, p. 107467, 2025.

[59] A. Li, M. Ogura, and N. Wakamiya, "Swarm shepherding using bearing-only measurements," *Philosophical Transactions of the Royal Society A*, vol. 383, no. 2289, p. 20240145, 2025.

[58] Y. Bai, P. T. T. Ngoc, H. D. Nguyen, D. L. Le, Q. H. Ha, K. Kai, Y. X. S. To, Y. Deng, J. Song, N. Wakamiya, H. Sato, and M. Ogura, "Swarm navigation of cyborg-insects in unknown obstructed soft terrain," *Nature Communications*, vol. 16, p. Article number: 221, 2025.

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[54] 多川純平, 小蔵正輝, and 杉本謙二, "Sparse Activity-Drivenネットワークにおける平均合意の収束性能解析," 計測自動制御学会論文集, vol. 60, no. 6, pp. 377–383, 2024.

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[39] R. Himo, M. Ogura, and N. Wakamiya, "Iterative algorithm for shepherding unresponsive sheep," *Mathematical Biosciences and Engineering*, vol. 19, no. 4, pp. 3509–3525, 2022.

[38] M. Kishida* and M. Ogura*, "Temporal deep unfolding for constrained nonlinear stochastic optimal controls," *IET Control Theory & Applications*, vol. 16, no. 2, pp. 139–150, 2022. (*equal contribution)

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[35] K. Sugimoto, T. Aihara, M. Ogura, and K. Hanada, "Gain scheduling for sampled-data state estimation over lossy networks," *Transactions of the Institute of Systems, Control and Information Engineers*, vol. 34, no. 11, pp. 287–293, 2021.

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[33] C. Zhao, M. Ogura, M. Kishida, and A. Yassine, "Optimal resource allocation for dynamic product development process via convex optimization," *Research in Engineering Design*, vol. 32, no. 1, pp. 71–90, 2021.

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[27] M. Kishida*, M. Ogura*, Y. Yoshida, and T. Wadayama, "Deep learning-based average consensus," *IEEE Access*, vol. 8, pp. 142 404–142 412, 2020. (*equal contribution)

[26] N. Masuda, V. M. Preciado, and M. Ogura, "Analysis of the susceptible-infected-susceptible epidemic dynamics in networks via the non-backtracking matrix," *IMA Journal of Applied Mathematics*, vol. 85, no. 2, pp. 214–230, 2020.

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[15] M. Wakaiki, M. Ogura, and J. P. Hespanha, "LQ-optimal sampled-data control under stochastic delays: gridding approach for stabilizability and detectability," *SIAM Journal on Control and Optimization*, vol. 56, no. 4, pp. 2634–2661, 2018.

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[12] M. Ogura and V. M. Preciado, "Optimal design of switched networks of positive linear systems via geometric programming," *IEEE Transactions on Control of Network Systems*, vol. 4, no. 2, pp. 213–222, 2017.

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[71] C. Zhao, K. Sakurama, and M. Ogura, "Optimization of Markovian switching one-way car-sharing networks via DC programming," in *23rd IFAC World Congress* (accepted), 2026.

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[68] H. Maeda and M. Ogura, "Modeling and analysis of locust swarms with OR35-deficient individuals," in *9th International Symposium on Swarm Behavior and Bio-Inspired Robotics 2025*, 2025.

[67] A. Sashie and M. Ogura, "Effects of differences in swimming speed on heterogeneous fish schools in a set-net environment," in *2025 SICE Festival with Annual Conference*, 2025, pp. 171–174.

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[63] W. Imahayashi, T. Tokugawa, K. Koitabashi, Y. Tsunoda, M. Ogura, and S. Muto, "Actual experiment using mobile robot in sheepdog system -guidance route design using traveling salesman problem-," in *SICE Festival 2024 with Annual Conference*, 2024, pp. 165–168.

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[61] A. Fujioka, M. Ogura, and N. Wakamiya, "Proposal and evaluation of pursuit formations based on cyclic pursuit dynamics," in *2023 International Symposium on Nonlinear Theory and its Applications*, 2023, pp. 30–33.

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[9] M. Ogura and C. F. Martin, "Stability of switching systems and generalized joint spectral radius," in *2013 European Control Conference*, 2013, pp. 3185–3190.

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Invited and Hourly Talks

[17] "Swarm control: From simple local rules to collective behavior," *Global Faculty Week, Kaunas University of Technology*, 2026.

[16] "From Local Interventions to Scalable System-Level Effects," *De La Salle University*, 2026.

[15] "Shepherding Dynamics: A Testbed for Nonlinear and Information-Limited Multi-Agent Control," *HKU-UM Workshop on Control Theory and Applications*, 2025.

[14] "Towards Operational and Scalable Control Strategies for Complex Systems," *Jiangnan University*, 2025.

[13] "Toward Control Theory for Complex Systems: From Epidemics to Swarms and Weather," *IDEC Institute, Hiroshima University*, 2025.

[12] "Shepherding as a paradigm for swarm control," *2024 IEEE International Conference on Cyborg and Bionic Systems*, 2024.

[11] "Enhancing Control System Design through Deep Unfolding: A Systematic Approach," *University of Hawaii at Manoa*, 2023.

[10] "Perspectives on Artificial Intelligence," *2nd Japanese-Canadian Frontiers of Science (JCFoS) Symposium*, 2023.

[9] "When deep unfolding meets control engineering," *37th International Technical Conference on Circuits\Systems, Computers and Communications*, 2022.

[8] "Optimization of positive linear systems via geometric programming," *Shenzhen University*, 2019.

[7] "Optimization of positive linear systems via geometric programming," *Guandong University of Technology*, 2019.

[6] "Synthesis of positive linear systems by geometric programming," *University of Hong Kong*, 2019.

[5] "Network epidemiology and control theory," *University of Hong Kong*, 2018.

[4] "Analysis and control of spreading processes over complex networks," *Washington State University*, 2017.

[3] "Analysis and control of spreading processes over complex networks," *Tokyo University of Agriculture and Technology*, 2016.

[2] "Stability analysis of switched linear systems with non-traditional switching signals," in *GRASP special seminar*, University of Pennsylvania, 2014.

[1] "Mean stability of switched linear systems," *Université Catholique de Louvain*, 2013.

TEACHING ACTIVITIES

Hiroshima University

- Reinforcement Learning (2025–present)
- Introduction to Artificial Intelligence (2024–present)
- Calculus II (2024–present)
- Artificial Intelligence and Machine Learning (2024)

Osaka University

- Basic Theory of Bio-networks (2021–present)
- Informatics for Engineering Science 1 (2020–present)
- Fundamentals of Information and Computer Sciences (2020–present)
- Bio-network Engineering (2020–present)

Nara Institute of Science and Technology

- Machine Learning and Intelligent Control (Spring 2019)
- Advanced Intelligent System Control (Spring 2017, 2018)

University of Pennsylvania

Co-lecturer:

- Convex Optimization in Systems and Control (Fall 2015)

Texas Tech University

Graduate Part-Time Instructor:

- Calculus II (Summer 2014, Spring 2014, Spring 2013)
- Calculus I (Summer 2013, Fall 2012)
- Trigonometry (Fall 2011)
- College Algebra (Fall 2013, Spring 2012)

Teaching Assistant:

- Advanced Calculus (Summer 2012)
- Linear Algebra (Summer 2012)
- Higher Mathematics for Engineers and Scientists I (Summer 2011)

Kyoto School of Computer Science

Lecturer:

- Control Engineering (Fall 2009, Fall 2008)
- Electrical Circuits (Spring 2008)
- Data Structures (Spring 2008)
- Numerical Analysis (Spring 2010, Spring 2009)

Kyoto University
Teaching Assistant:

- Modern Control Theory (Fall 2009, Fall 2008)

PROFESSIONAL SERVICE

- Program Committee Member, Sixth International Symposium on Complex Systems
- Associate Editor, 2026 SICE International Symposium on Control Systems
- Organized & Invited Sessions Chair, SICE Festival 2026 with Annual Conference
- Program Committee Member, 14th International Conference on Complex Networks and their Applications
- Associate Editor, The Journal of Engineering (2025–present)
- Technical Comittee Member, Technical Comittee on Fully Actuated System Theory and Applications (2025–present)
- International Program Committee Secretary, 8th IFAC Conference on Nonlinear Model Predictive Control
- Publication Co-Chair, 8th IFAC Conference on Nonlinear Model Predictive Control
- Program Committee Member, 13th International Conference on Complex Networks and their Applications
- Associate Editor, 10th Indian Control Conference
- External Reviewer, Research Grants Council of Hong Kong (2023, 2024, 2026)
- Reviewer, French National Research Agency (2023)
- Program Committee member, 12th International Conference on Complex Networks and their Applications
- Registration Chair, 10th IFAC Symposium on Robust Control Design
- International Advisory Committee Member, 2022 International Conference on Information Technology
- Associate Editor, IET Control Theory & Applications (2022–present)
- Associate Editor, Franklin Open (2022–present)
- Program Committee member, 11th International Conference on Complex Networks and their Applications
- Program Committee member, 10t International Conference on Complex Networks and their Applications
- Technical Program Comittee Member, 7th IEEE International Conference on Network Softwarization
- Associate Editor, Journal of The Franklin Institute（2020–present）
- Program & Steering Committee Member, 52nd ISCIE International Symposium on Stochastic Systems Theory and Its Applications
- Local Arrangements Vice Chair, SICE Annual Conference 2018
- Technical Comittee Member, Technical Committee 1.5. Networked Systems, International Federation of Automatic Control (2015–present).
- Associate Editor, 5th IFAC Workshop on Distributed Estimation and Control in Networked Systems

Journal Reviewer

Advanced Robotics; Annual Reviews in Control; Automatica; Applied Mathematics and Computation; Applied Network Science; Applied Soft Computing; Asian Journal of Control; Chaos, Solitons and Fractals; Computer Communications; Developments in the Built Environment; Discrete Mathematics; Engineering Applications of Artificial Intelligence; Entropy; European Journal of Control European Physical Journal B; Expert Systems With Applications; Foundations of Computational Mathematics; IEEE Access; IEEE/ACM Transactions on Computational Biology and Bioinformatics; IEEE/CAA Journal of Automatica Sinica; IEEE Circuits and Systems Magazine; IEEE Control Systems Letters; IEEE Intelligent Systems; IEEE Internet of Things Journal; IEEE Open Journal of Vehicular Technology; IEEE Signal Processing Letters; IEEE Transactions on Automatic Control; IEEE Transactions on Automation Science and Engineering; IEEE Transactions on Big Data; IEEE Transactions on Circuits and Systems I: Regular Papers; IEEE Transactions on Circuits and Systems II: Express Briefs; IEEE Transactions on Cognitive and Developmental Systems; IEEE Transactions on Computational Social Systems; IEEE Transactions on Control of Network Systems; IEEE Transactions on Cybernetics; IEEE Transactions on Fuzzy Systems; IEEE Transactions on Industrial Electronics; IEEE Transactions on Industrial Informatics; IEEE Transactions on Network Science and Engineering; IEEE Transactions on Neural Networks and Learning Systems; IEEE Transactions on Signal and Information Processing over Networks; IEEE Transactions on Signal Processing; IEEE Transactions on Systems, Man and Cybernetics: Systems; IEEE Transactions on Pattern Analysis and Machine Intelligence; IEEE Transactions on Power Systems; IEEE Transactions on Mobile Computing; IET Control Theory & Applications; International Journal of Systems Science; International Journal of Robust and Nonlinear Control; Internet Technology Letters; Journal of Aerospace Engineering; Journal of Biological Dynamics; Journal of Building Engineering; Journal of Computational Science; e Journal of Systems Science and Complexity; Journal of the Franklin Institute; Mathematics and Computers in Simulation; Mathematics of Control, Signals, and Systems; National Science Review; Neural Networks; Neurocomputing; Nonlinear Analysis: Hybrid Systems; Nonlinear Dynamics; Operations Research and Decisions; Optimal Control, Applications and Methods; Philosophical Transactions of the Royal Society A; Physica A; Physics Letters A; PLOS Computational Biology; Science China Information Sciences; SIAM Journal on Control and Optimization; SICE Journal of Control, Measurement, and System Integration; Stochastics and Dynamics; Systems and Control Letters; Research in Engineering Design.

Masaki Ogura, October 2026